

SMART AGRICULTURE And AGRICULTURAL AUTOMATION



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Agriculture is an important sector in a country's economy. It fulfils the demand for food of a nation. Sensors play an important role in this sector. These make agricultural work easier and increase agricultural productivity.

Agriculture is harnessing computer hardware, software, electronic instruments and control systems. Some related areas are agronomy, horticulture, veterinary medicine and food processing. Integrating sensors, electronics, control and power systems, and ICT engineering for agriculture is a key enabler for delivering improved food supply and sustainable energy production without increased burden on the limited fertile land bank. This exploration of non-traditional technologies to improve global farming is extremely time critical.

Smart agriculture, also known as precision agriculture, allows farmers to maximise yields using minimum resources, such as water, fertilisers and seeds. By deploying sensors and mapping fields,

they can understand their micro-scale, conserve resources and reduce impact on the environment.

Smart agriculture has its roots going back to 1980s when global positioning system (GPS) capability became accessible for civilian use. Once farmers were able to accurately map their crop yields, they could monitor and apply fertiliser and weed treatments only to the areas that required it.

During 1990s, early precision agriculture users adopted crop yield monitoring to generate fertiliser and pH correction recommendations. As more variables could be measured and entered the crop model, more accurate recommendations for fertiliser application, watering and even peak yield harvesting could be made.

Agricultural sensors

Several sensing technologies are used in precision agriculture. These provide data that assist farmers to monitor and optimise crops as well as adapt to changing environmental factors. Some sensing technologies used in

Precision agriculture allows farmers to collect and process data in real time and make the best decisions with regards to planting, fertilising and harvesting crops